

Question 1

(a)

Total arrangements: $\binom{5}{4} \times 4! = 5 \times 24 = 120$.

Probability per arrangement: $\frac{1}{120}$.

$$\boxed{120}$$

$$\boxed{\frac{1}{120}}$$

(b)

Probability Alice and Dara together: $4 \times \binom{3}{2} \times 2! \times 2/120 = 48/120 = 2/5$.

So not together: $1 - 2/5 = 3/5$.

$$\boxed{\frac{3}{5}}$$

(c)

Circular table, 4 people with 5 seats.

Total circular arrangements: $(5 - 1)! = 24$. (have extra seat)

Arrange 4 people in remaining 4 seats: $4! = 24$

Circular symmetry, divide by 5 $\frac{5 \times 24}{5} = 24$

Remaining 2 seats: $2! = 2$

Number of circular arrangements: $(4 - 1)! = 6$. So total: $6 \times 2 = 12$ Arrangement where they are not adjacent: $24 - 12 = 12$ Probability: $\frac{12}{24} = \frac{1}{2}$

$$\boxed{\frac{1}{2}}$$

Question 2

(a)

$$P(\text{at least one hit}) = 1 - (0.3)^3 = 1 - 0.027 = 0.973.$$

0.973

(b)

$$P(\text{first fails and second succeeds}) = 0.027 \times 0.973 = 0.026271.$$

0.026271

(c)

$$P(\text{interception}) = 0.973 + 0.026271 = 0.999271.$$

$$E[X - \text{intercept}] = (3 \times 0.973 + 6 \times 0.026271)/0.999271 \approx 3.0789 \approx 3.08.$$

$$\text{Var}(X) \approx 0.23.$$

3.08

0.23

Question 3

(a)

$$P(\text{same color}) = \frac{\binom{n}{2} + \binom{m}{2}}{\binom{n+m}{2}} = \frac{n(n-1) + m(m-1)}{(n+m)(n+m-1)}.$$

$$\boxed{\frac{n(n-1) + m(m-1)}{(n+m)(n+m-1)}}$$

(b)

$$P(\text{same color}) = \left(\frac{n}{n+m}\right)^2 + \left(\frac{m}{n+m}\right)^2 = \frac{n^2+m^2}{(n+m)^2}.$$

$$\boxed{\frac{n^2 + m^2}{(n+m)^2}}$$

(c)

(b) is larger because $P_b - P_a = \frac{2nm}{(n+m)^2(n+m-1)} > 0$ for $n, m > 0$.

$\boxed{\text{(b) is larger}}$

Question 4

$$P(\text{smoker} \text{ --- ectopic}) = \frac{2p \times 0.32}{2p \times 0.32 + p \times 0.68} = \frac{0.64}{1.32} = \frac{16}{33} \approx 48.48\%.$$

48.48%

Question 5

$P(\text{first spade} \mid \text{second and third spades}) = \frac{11}{50}$ (since 11 spades left out of 50 cards).

$$\boxed{\frac{11}{50}}$$

Question 6

(a)

$$P(\text{upstairs}) = 0.7 \times 0.6 + 0.5 \times 0.4 = 0.42 + 0.20 = 0.62.$$

$$\boxed{0.62}$$

(b)

$$P(\text{dad} \text{ --- downstairs}) = \frac{0.5 \times 0.4}{0.3 \times 0.6 + 0.5 \times 0.4} = \frac{0.20}{0.38} = \frac{10}{19}.$$

$$\boxed{\frac{10}{19}}$$

Question 7

(a)

$$P(\max = k) = \frac{2k-1}{36} \text{ for } k = 1, 2, 3, 4, 5, 6.$$

$$P(M = k) = \frac{2k-1}{36} \quad k = 1, 2, 3, 4, 5, 6$$

(b)

$$P(\min = k) = \frac{13-2k}{36} \text{ for } k = 1, 2, 3, 4, 5, 6.$$

$$P(N = k) = \frac{13-2k}{36} \quad k = 1, 2, 3, 4, 5, 6$$

(c)

$$P(\text{sum} = k) = \begin{cases} 1/36 & k = 2, 12 \\ 2/36 & k = 3, 11 \\ 3/36 & k = 4, 10 \\ 4/36 & k = 5, 9 \\ 5/36 & k = 6, 8 \\ 6/36 & k = 7 \end{cases}$$

$$P(S = k) = \begin{cases} \frac{1}{36} & k = 2, 12 \\ \frac{2}{36} & k = 3, 11 \\ \frac{3}{36} & k = 4, 10 \\ \frac{4}{36} & k = 5, 9 \\ \frac{5}{36} & k = 6, 8 \\ \frac{6}{36} & k = 7 \end{cases}$$

(d)

$$P(\text{diff} = k) = \frac{6-|k|}{36} \text{ for } k = -5, -4, \dots, 5.$$

$$P(D = k) = \frac{6-|k|}{36} \quad k = -5, -4, \dots, 5$$

Question 8

(a)

$$\sum_{i=1}^6 ci = c \times 21 = 1 \implies c = 1/21.$$

$$\boxed{\frac{1}{21}}$$

(b)

$$P(X=i) = \frac{i}{21} \text{ for } i = 1, 2, 3, 4, 5, 6.$$

$$\boxed{P(X = i) = \frac{i}{21} \quad i = 1, 2, 3, 4, 5, 6}$$

(c)

$$E[X] = \frac{1}{21} \sum i^2 = \frac{91}{21} = \frac{13}{3}. \quad \text{Var}(X) = 21 - \left(\frac{13}{3}\right)^2 = \frac{20}{9}.$$

$$\boxed{E[X] = \frac{13}{3}}$$

$$\boxed{\text{Var}(X) = \frac{20}{9}}$$

(d)

$$E[(4X + 1)^2] = 16 \times 21 + 8 \times \frac{13}{3} + 1 = \frac{1115}{3}. \quad \text{Var}[5 - 2X] = 4 \times \frac{20}{9} = \frac{80}{9}.$$

$$\boxed{E[(4X + 1)^2] = \frac{1115}{3}}$$

$$\boxed{\text{Var}[5 - 2X] = \frac{80}{9}}$$

Question 9

$P(Y \geq 1)$ for one pack: $1 - (0.99)^{20} - 20 \times 0.01 \times (0.99)^{19} \approx 0.016859$.

Expected refund for 100 packs: $100 \times 25 \times 0.016859 \approx 42.15$.

42.15

Question 10

(a)

Rate in 6 min: $\lambda t = 4$. $P(X=0) = e^{-4}$.

$$e^{-4}$$

(b)

$P(X \geq 3) = 1 - e^{-4}(1 + 4 + 8) = 1 - 13e^{-4}$.

$$1 - 13e^{-4}$$

(c)

Interarrival $T \sim \text{Exp}(\lambda = 2/3)$. $P(T > 15) = e^{-(2/3) \times 15} = e^{-10}$.

$$e^{-10}$$